

Agent: host_agent

User: Show me all the playing fields within Exeter.

Agent: planning_agent

The area is Exeter, so there will be 1 entry for Exeter. The entities to search for are playing fields, and we need to find as many entries as possible within Exeter.

output steps: ["Find Exeter 1 area", "Search for playing fields in Exeter as many search results"]

Tool Call: send_message

- target: named_area
- task_description: Find the area polygon for Exeter.

Agent: named_area

Tool Call: call_os_ngd

- filters: None
- bbox: None
- polygon_or_point: True
- street_address: None
- filename: exeter_polygon

Tool call result: Search results have been found. Artifact generated is : exeter_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter., count of records fetched : 98.

Tool Call: send_message

- target: data_analysis_agent
- task_description: Filter the artifact 'exeter_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon'.

Agent: data_analysis_agent

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    mask = (
        (gdf['name1_text'].str.lower() == 'exeter') &
        (
            (gdf['descriptiongroup'].str.lower().isin(['settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town']))
        )
    )
    filtered = gdf[mask]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter area."
    artifact_name = 'filtered_exeter_polygon'
    artifact_description = 'Polygon(s) representing the specific area for Exeter, filtered by name and settlement/city/town type.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool Code Executor

```
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    mask = (
        (gdf['name1_text'].str.lower() == 'exeter') &
        (
            (gdf['descriptiongroup'].str.lower().isin(['settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town']))
        )
    )
    filtered = gdf[mask]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter area."
    artifact_name = 'filtered_exeter_polygon'
    artifact_description = 'Polygon(s) representing the specific area for Exeter, filtered by name and settlement/city/town type.'
    return [summary, artifact_name, artifact_description, filtered]
```

```
# Call the function with the provided data list
filter_exeter_polygon([exeter_polygon])
```

Tool Code Executor

```
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    mask = (
        (gdf['name1_text'].str.lower() == 'exeter') &
        (
            (gdf['descriptiongroup'].str.lower().isin(['settlement'])) |
            (gdf['description'].str.lower().isin(['city', 'town']))
        )
    )
    filtered = gdf[mask]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter area."
    artifact_name = 'filtered_exeter_polygon'
    artifact_description = 'Polygon(s) representing the specific area for Exeter, filtered by name and settlement/city/town type.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool code result: ['Filtered to 1 polygon(s) for Exeter area.',]

Output: Filtered to 1 polygon for the Exeter area. The artifact 'filtered_exeter_polygon' contains the polygon(s) representing the specific area for Exeter, filtered by name and settlement/city/town type.

Agent: named_area

Output: The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_polygon'. This artifact contains the polygon(s) representing the area for Exeter, filtered by name and settlement/city/town type.

Tool Call: send_message

- target: land_use_features
- task_description: Search for playing fields within the area defined by the artifact 'filtered_exeter_polygon'.

Agent: land_use_features

Tool Call: call_os_ngd

- filters: ['Playing Field']
- bbox: filtered_exeter_polygon
- polygon_or_point: True
- street_address: False
- filename: playing_fields_exeter

Tool call result: Multiple search results have been found in multiple datasets. Artifacts generated are : ['landuse_playing_fields_exeter'], Descriptions are : ["A geopandas dataframe containing land use data with filters and bbox applied for the query Search for playing fields within the area defined by the artifact 'filtered_exeter_polygon'. using filters ['Playing Field']. Further Name filtering is available for this"], counts of records fetched are : [6].

Output: I found 6 playing fields within the area defined by the artifact 'filtered_exeter_polygon'. The data is saved in the artifact named 'landuse_playing_fields_exeter'.

Tool Call: send_message

- target: plotting_agent
- task_description: Plot the playing fields from the artifact 'landuse_playing_fields_exeter' within the area of Exeter from 'filtered_exeter_polygon'. Add a summary of the results.

Agent: plotting_agent

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def plot_playing_fields_within_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack the data
    playing_fields = data[0]
```

```

exeter_poly = data[1]

# Ensure both are in the same CRS (EPSG:4326 for folium)
playing_fields = playing_fields.to_crs(epsg=4326)
exeter_poly = exeter_poly.to_crs(epsg=4326)

# Get the Exeter city polygon
exeter_geom = exeter_poly.iloc[0].geometry

# Filter playing fields that intersect with Exeter polygon
within_exeter = playing_fields[playing_fields.intersects(exeter_geom)]

# Create a folium map centered on Exeter
centroid = exeter_geom.centroid
m = folium.Map(location=[centroid.y, centroid.x], zoom_start=12)

# Add Exeter boundary
folium.GeoJson(mapping(exeter_geom), name='Exeter Boundary', style_function=lambda x: {'color': 'blue', 'weight': 2, 'fill': False}).add_to(m)

# Add playing fields
for _, row in within_exeter.iterrows():
    folium.GeoJson(
        mapping(row.geometry),
        name=row.get('name1_text', 'Playing Field'),
        style_function=lambda x: {'color': 'green', 'weight': 2, 'fill': True, 'fillOpacity': 0.4}
    ).add_to(m)
    # Add popup with name if available
    popup_text = row['name1_text'] if row['name1_text'] else 'Playing Field'
    folium.Popup(popup_text).add_to(folium.GeoJson(mapping(row.geometry)))

# Save map
map_filename = 'playing_fields_within_exeter.html'
m.save(map_filename)

# Prepare summary
summary = f"There are {len(within_exeter)} playing fields within the Exeter city boundary.\n"
if len(within_exeter) > 0:
    summary += "First 5 playing fields:\n"
    for i, row in within_exeter.head(5).iterrows():
        summary += f"- OSID: {row['osid']], Area: {row['geometry_area_m2']:.2f} m , Name: {row['name1_text'] if row['name1_text'] else 'N/A'}\n"
else:
    summary += "No playing fields found within Exeter.\n"

artifact_description = "A folium map showing all playing fields (landuse_playing_fields_exeter) within the Exeter city boundary (filtered_exeter_polygon),

return [summary, map_filename, artifact_description, map_filename]

```

Tool code result: There are 6 playing fields within the Exeter city boundary. First 5 playing fields: - OSID: 6e11881d-2c89-4e6f-a506-274315538527, Area: 145675.16 m , Name: N/A - OSID: 81aad8de-203a-48a0-927d-779798e2a3d8, Area: 5098.40 m , Name: N/A - OSID: 5b6b234e-3da8-47ef-95aa-1ca6b73bb261, Area: 9849.09 m , Name: N/A - OSID: d3a33762-19b2-4805-9c56-6c2a0b99ca29, Area: 133914.63 m , Name: N/A - OSID: 17a457f6-ea11-4568-b195-ce1ba5899cbc, Area: 1225.47 m , Name: N/A

An artifact has been generated with the name: playing_fields_within_exeter.html

Description: A folium map showing all playing fields (landuse_playing_fields_exeter) within the Exeter city boundary (filtered_exeter_polygon), with the Exeter boundary in blue and playing fields in green.

Artifact: playing_fields_within_exeter.html

Agent: host_agent

There are 6 playing fields within the Exeter city boundary. These have been identified and mapped using the latest land use data. Here's a summary of the results:

- The playing fields are distributed across Exeter, with areas ranging from about 1,225 m to over 145,000 m .
- The map highlights the Exeter city boundary in blue and all playing fields in green.

You can view the interactive map in the artifact: playing_fields_within_exeter.html, which shows the exact locations and extents of all playing fields within Exeter. If you need more details about any specific playing field, let me know!